

Aruvi — the external-provider seam

Every vendor-shaped dependency sits behind a named interface. This sheet shows each slot, what fills it today, and what a production partner is being asked to supply.

ARUVI'S OWN CODE — RETAINED, NOT REBUILT

Answers Section 3 of the Technology Partnership & Production Roadmap: what can be retained, and what must be replaced.

web/ <i>Next.js</i> The teacher-facing app. Mobile-first and responsive. Speaks HTTP to the API and to nothing else.	api/ <i>FastAPI service</i> The service layer. Every provider in the product is chosen in ONE block: main.py, lines 79-227.	aruvi_core/ <i>the engine</i> Lesson generation, period allocation, canonical view model, five subject plugins. Pure Python, no UI.	genon/ <i>authoring pipeline</i> Runs offline to build the lesson library. Not part of the live service, and not part of the handover. Unrelated to aruvi_core/genon/, which is the live serve engine.
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No vendor name appears anywhere above this line.

aruvi_core/ports.py — 22 named interfaces

The engine depends on these and never on a provider. Swapping a vendor is one new adapter class plus one line in api/main.py. Every one of the 22 is now honoured — no caller reaches past a port.

THE SEAM

EXTERNAL PROVIDERS — THE SLOTS A PRODUCTION PARTNER FILLS

Four have working plugs, two are stubs, two are reserved for later — and no seam is bypassed.

WORKING PLUG • FILES Database / teacher state 12 × ...Repository TODAY Twelve file adapters under data/cloud/state/. Every method already takes tenant_id and user_id (teaching state adds year_id) and paths are keyed to match — so tenant isolation is already in the shape of the data, not a later retrofit. PARTNER SUPPLIES Managed Postgres (e.g. Supabase) behind the same twelve ports, plus row-level security.	STUB • THE SECURITY GAP Authentication AuthProvider TODAY HeaderAuthProvider: the user ID arrives in an X-Aruvi-User header and there is no password. Development only. Exactly one caller, _current_identity(). PARTNER SUPPLIES A real identity provider with token verification. Replaces one class and one line.	STUB • MANUAL TODAY Payments BillingProvider TODAY ManualBillingProvider — the founder grants entitlement by CLI. Everything around it is built: entitlement scopes, invoice series, consent record, priced cart. PARTNER SUPPLIES An Indian gateway plus a webhook route, and advice on Play / App Store fees across web and mobile.	WORKING PLUG • REAL SMTP Transactional email Notifier TODAY SmtplibNotifier sends genuinely once credentials are set in the environment; a file outbox stands in otherwise. The choice is made once, at the seam, on startup. PARTNER SUPPLIES Optional — a deliverability provider if volume warrants it. Nothing here is blocking.
WORKING PLUG • PORT HONoured Content / object store Storage TODAY LocalStorage serves the whole content tree — lesson library, mappings, glossaries, norms, the user agreement. The port was widened to answer "which canonicals exist?", which is why it used to be bypassed. No runtime filesystem call against the library survives. PARTNER SUPPLIES S3 / GCS / Supabase Storage behind the same port — one adapter class, no caller changes.	RESERVED • NONE AT RUNTIME AI / LLM LLMClient, OutputCache TODAY Not wired, deliberately. The live service calls no AI at all — it serves pre-authored plans by deterministic selection. There is no AI key in production to leak. PARTNER SUPPLIES Nothing now. An adapter only if live generation is later wanted.	RESERVED • UNUSED Background jobs JobQueue TODAY Declared and unused. Nothing in the product currently needs asynchronous work; exports render inline. PARTNER SUPPLIES Nothing now. Relevant only if batch work later moves server-side.	WORKING PLUG • BUILT Data rights (DPDP • Apple 5.1.1(v)) DataRightsService TODAY Export walks every state store into a single document; erase returns a signed receipt. Neither is ever gated on subscription state. PARTNER SUPPLIES Nothing of its own — it travels with the database adapter.

Config, not code

Every provider setting is an environment variable in api/config.py. No secret is committed to the repository.

The contracts are tested

43 test files exercise the ports — including one that runs the runtime against a store with no filesystem at all — so a swapped adapter is verifiable rather than hopeful.

Ownership

Cloud, database, domain, store and gateway accounts to be created in Aruvi's own name. The partner receives access, not ownership.